



Lab 4: Analyzing Digital Transmissions using Wireshark

Introduction to Digital Transmissions

In modern computer networks, digital transmission is the foundation of reliable communication. Data is transmitted as digital signals over network cables or wireless channels. The efficiency of this transmission depends on multiple factors, including transmission modes, encoding schemes, bandwidth usage, data rate, signal rate, and error detection mechanisms. These elements collectively determine how efficiently and accurately data moves across a network.

[Wireshark](#), a powerful packet analyzer, allows us to observe real network traffic, measure transmission speeds, and analyze error detection mechanisms. By capturing packets in real time, we can gain valuable insights into how digital transmission works in practice.

Transmission Modes: Parallel vs. Serial

Transmission modes define how data moves from one device to another. In parallel transmission, multiple bits are sent simultaneously over multiple channels. This method is used in short-distance communication, such as transferring data inside a computer between the processor and memory. Since each bit travels on a separate wire, parallel transmission can be fast but requires more hardware and is prone to synchronization issues over long distances.

In contrast, serial transmission sends bits one at a time over a single channel. This method is widely used in networking technologies such as Ethernet, Wi-Fi, and USB, where data must travel efficiently over long distances. Serial transmission is slower per cycle than parallel transmission but is more reliable for network communication. In this lab, we will analyze ICMP (ping) and TCP packet sequences in Wireshark to observe how serial transmission works in real networks.

Data Rate vs. Signal Rate

A fundamental measure of transmission efficiency is the data rate, also called the bit rate, which refers to the number of bits transmitted per second (bps). However, data is not always transmitted as individual bits; instead, multiple bits can be encoded within a single signal change. The signal rate, or [baud rate](#), refers to the number of signal changes per second. If a transmission system encodes multiple bits per signal change, the baud rate will be lower than the bit rate.

$$\text{Signal Rate (Baud)} = \text{Bit Rate (bps)} / \text{Bits per symbol (r)}$$



For example, if a system transmits data at 10 Mbps while encoding two bits per signal change, the baud rate will be 5 Mbaud (**note that the transmitted data was in Megabits and thus the final result is Megabaud**). Understanding this relationship is crucial for optimizing bandwidth usage in digital networks. In this lab, we will capture network traffic, extract real-time timestamps, and compute the data rate versus signal rate to compare theoretical and observed values.

Bandwidth and Nyquist Theorem

Bandwidth refers to the data transfer rate of a network. In digital transmission, the amount of bandwidth needed depends on the data rate and the encoding scheme used. The **Nyquist Theorem** states that in order to accurately reconstruct a signal, the sampling rate must be at least twice the highest frequency present in the signal:

Sampling Rate $\geq 2 \times$ Maximum Frequency

For example, human voice signals typically have a maximum frequency of 4 kHz. To digitize voice signals accurately, a minimum sampling rate of 8 kHz is required. This principle is applied in various technologies, such as telephony and audio processing, to ensure high-quality signal transmission.

In this lab, we will use Wireshark to compare bandwidth consumption for different types of network traffic, such as web browsing, video streaming, and file downloads. By analyzing real-world data, we will determine how different applications affect bandwidth usage and network performance.

Error Detection in Digital Transmission

During digital transmission, errors can occur due to noise, interference, or signal degradation. To ensure data integrity, networks use error detection techniques embedded within network protocols. One of the most common error detection methods is the **Frame Check Sequence (FCS)**, found in Ethernet frames. This sequence allows the receiver to verify whether a packet has been corrupted during transmission.

Another widely used technique is the **checksum**, which is included in TCP/IP packets. The checksum algorithm calculates a unique value based on the contents of the packet. When the receiver receives the packet, it performs the same calculation and compares the result. If there is a mismatch, the packet is considered corrupted and may be discarded or retransmitted.

A more advanced error detection technique is the **Cyclic Redundancy Check (CRC)**, which is used in various network and storage systems to detect errors in data. Unlike simple checksums, CRC uses polynomial division to generate a more robust error-checking value.



In this lab, we will analyze real network packets in Wireshark to observe these error detection mechanisms in action. We will also identify packets with checksum errors and investigate whether errors occur more frequently in wired or wireless network connections.

Lab Tasks

Note: All tasks must be documented in a single report. Include screenshots of every step (description is not as necessary as the screenshots).

CLO1: Explain the fundamental principles, architectures, and communication models of computer networks, highlighting the roles of different layers in end-to-end data exchange.

Before starting, ensure that a notepad or sticky note containing your name, roll number, section, and lab number is clearly visible. All screenshots must include this note, otherwise, they will not be considered.

Task 1: Capturing and Understanding Packet Transmission

The first step in analyzing digital transmission is to capture network packets and examine their transmission characteristics. You will capture **ICMP (ping) packets** to observe how data is transmitted serially over the network.

For this you will:

Open **Wireshark** and select the **active network interface** (Ethernet or Wi-Fi). Start a packet capture session by clicking the **Start Capture** button. Open the command prompt on your system and run this: `ping -n 5 www.google.com`. Which will send five **ICMP echo request** packets to Google's server. After the ping operation completes, **stop the packet capture** in Wireshark. In the Wireshark **filter bar**, enter the `icmp` filter to isolate ICMP packets.

Select an **ICMP request packet** and examine the **Frame Details**. Record the following information on a document:

1. Packet arrival time (timestamp) [of any packet]
2. Frame length (bytes) [of any packet]
3. Time difference between consecutive packets (inter-packet delay) [first and last request]

Based off this information, answer the following questions:

Q1. How frequently are packets being sent?



Q2. What is the **average packet size**?

Q3. What transmission mode is used?

Task 2: Data Rate vs. Signal Rate Calculation

Capture **TCP packet transmissions** to analyze data rates and calculate the signal rate (baud rate).

For this:

Open Wireshark and start a **new packet capture**. Download a large file (e.g., from **Google Drive, OneDrive, or a website**) to generate sustained traffic. Stop the capture after a few seconds. Apply the *TCP* filter to display only TCP packets. Navigate to **Statistics** then go to **I/O Graph** and observe the transmission pattern. Identify the total number of bytes transmitted and the duration of the transmission on the same document as task 1 (under a new heading for task 2). [you can also view it on Conversations]

Calculate the following:

1. Data Rate (Bit Rate) = Total bits transmitted / Transmission duration
2. Packet Transmission Rate (Packets/sec) = Total packets captured / Transmission duration
3. Baud Rate (Signal Rate) using the formula given in the manual at the start. (hint: to identify the bits per symbol, you must run the command *netsh wlan show interfaces* on your system. Then find the **modulation scheme** for your network. You may use AI for only this question. Attach a screenshot of your prompt and the result you got in your submission.)

Based off this information, answer the following questions:

Q1. How does the observed **data rate** compare to the **expected network speed**? (hint: you will need to check your connection speed to see the expected network speed)

Q2. If multiple bits per symbol are used, how does this affect the baud rate?

Task 3: Measuring Bandwidth Usage for Different Traffic Types

Investigate how different applications consume bandwidth by analyzing **web browsing and video streaming**.

For this:

Open Wireshark and start a new capture session.



Perform the following network activities **one at a time** while capturing packets:

- **Browse a website** (e.g., search on Google).
- **Watch a video** (YouTube, Netflix, or Twitch).

Stop the capture and apply the appropriate filters. For web browsing use the TCP filter and for video streaming use the UDP filter (hint: you are going to need a filter like frame contains “google” or tcp contains “google”).

Open **Statistics** then go to **Conversations** and examine the total **Bytes and Packets exchanged** for each activity. Compare the **bandwidth usage** of web browsing and video streaming by answering the following questions.

Q1. Which type of traffic generates the **most packets** and consumes the highest bandwidth? Mathematically prove it using the given formula.

$$\text{Bandwidth (bps)} = \text{Total Bytes} \times 8 / \text{Time (seconds)}$$

Q2. Does streaming use **larger or smaller packets** than web browsing (based on your observation)?

Q3. How does UDP (used in video streaming) compare to TCP (used in web browsing)?

Task 4: Identifying Error Detection Techniques in Packet Headers

Many **error detection mechanisms** such as **checksums**, **Frame Check Sequence (FCS)**, and **Cyclic Redundancy Check (CRC)** exist. We will view one simple mechanism in this lab.

For this:

Open Wireshark and start capturing network traffic. Select a **TCP segment** from the captured packets and expand the **Frame Details** or **expand TCP**.

Locate and examine the **TCP Checksum** which verifies data integrity in TCP packets. (hint use: tcp.checksum filter)

Apply the *tcp.analysis.flags* filter to identify packets with checksum errors. Compare error rates in **wired vs. wireless** network captures to observe any differences (you can connect an ethernet cable to your laptop/PC to test this).

Based off your observations answer the following questions:

Q1. How frequently do checksum errors occur in **wired vs. wireless networks**?



CL3001 - Computer Networks Lab

Instructors: Ms. Zoya Mahboob

Lab 4 - Analyzing Digital Transmissions using Wireshark

Q2. Why are **Wi-Fi packets more prone to errors** compared to Ethernet?

Submission Guidelines (5 Marks):

- Submit a single **PDF/Word** document containing all tasks to the GCR. **(DO NOT SUBMIT THE WIRESHARK FILES)**
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **L_i231234_Lab1**.
- The unethical usage of Generative AI is strictly prohibited and will result in 0 marks.
- **In case a demo is conducted, there will be deductions based on your answers to the questions asked.**

NOTE: Screenshots where your name, roll number, section AND lab number are NOT visible, will automatically be given 0/not considered valid.

Sample screenshot:

